USC Dornsife
Dana and David Dornsife
College of Letters, Arts and Sciences
Spatial Sciences Institute

SSCI 683: Principles of Spatial Data Analysis

Units: 4

Term—Day—Time: Spring 2026, Fridays, 9 – 11:50 a.m.

Location: DMC 108

Instructor: An-Min Wu, Ph.D.
Office: AHF B55B
Office Hours: Monday 2 - 3 p.m. and Tuesday 3 - 4 p.m. I am also often available to meet Mondays to Fridays between 9 a.m. – 5 p.m. outside the office hours; contact me to make the meeting appointment.
Contact Info: anminwu@usc.edu

Library Help: Andrzej (Andy) Rutkowski
Office: LIPA B40-A
Office Hours: By appointment
Contact Info: arutkows@usc.edu

IT Help: Spatial Support
Contact Info: spatial_support@usc.edu

Course Scope and Purpose

This course explores theoretical foundations, methods, techniques, and software systems for spatial data analysis. The course aims to provide students with the fundamental knowledge and hands-on skills necessary to investigate research questions, with a focus on population and health problems, using current spatial analytic approaches. Including fundamental spatial concepts and the core technical components, this course intends to explore a broad range of the principles of spatial analysis methods, including exploratory spatial data analysis, remote sensing, spatial statistics, and network and locational analysis. The latest research in a variety of topics related to population, health, and place that are central to spatial analysis are also examined. Students will gain an in-depth understanding and hands-on experience in the ways to explore a variety of applications through a combination of homework, presentations, and projects. Students will learn about the wide variety of geospatial data and analytical tools available, including how to find relevant data and transform it as needed so that it can be used for solving specific socio-economic challenges and problems.

Learning Objectives

On completion of this course, students should be able to:

- Describe key theoretical concepts of spatial analysis approaches.
- Evaluate spatial analysis approaches and techniques for working on research that investigates places and their roles in shaping environmental exposures, health-related impacts and outcomes, and the efficacy of health care delivery systems.
- Analyze spatial problems by applying appropriate and relevant spatial analysis techniques.

Prerequisite(s): None.
Co-Requisite (s): Students must be enrolled in an existing USC PhD program
Concurrent Enrollment: None
Recommended Preparation: Some experience in GIS software or GIS coursework (e.g. SSCI 581) is strongly recommended; contact the instructor if you have no prior experience in GIS.

Class Conduct

Harassment, sexual misconduct, interpersonal violence, and stalking are not tolerated by the university. All faculty and most staff are considered Responsible Employees by the university and must forward all information they receive about these types of situations to the Title IX Coordinator. The Title IX Coordinator is responsible for assisting students with supportive accommodations, including academic accommodations, as well as investigating these incidents if the reporting student wants an investigation. The Title IX office is also responsible for coordinating supportive measures for transgender and nonbinary students such as faculty notifications, and more. If you need supportive accommodations you may contact the Title IX Coordinator directly (titleix@usc.edu or 213-821-8298) without sharing any personal information with me. If you would like to speak with a confidential counselor, Relationship and

This content is protected and may not be shared, uploaded, or distributed.

Sexual Violence Prevention Services (RSVP) provides 24/7 confidential support for students (213-740-9355 (WELL); press 0 after hours)

SSI Policy on the Creation of Original Work and Use of Generative AI – All students are expected to submit assignments that represent their own original work, and that have been prepared specifically for the course or section for which they have been submitted. Students may not have another person or entity complete any substantive portion of an assignment or reuse work prepare for courses without obtaining written permission from the instructor(s). Developing strong competencies in research, writing, and the technical execution of geospatial technologies are foundational to SSI academic programs that are designed to prepare you for success in the workplace. Therefore, using generative AI tools – unless explicitly specified otherwise – is strictly prohibited in this course, will be identified as plagiarism, and will be reported to the Office of Academic Integrity.

Diversity and Inclusion – It is my intent that students from all diverse backgrounds and perspectives be well served by this course, that students' learning needs be addressed both in and out of class, and that the diversity that students bring to this class be viewed as a resource, strength and benefit. It is my intent to present materials and activities that are respectful to everyone, and you are also expected to respect of others regardless of their race, ethnicity, gender identity and expressions, cultural beliefs, religion, sexual orientation, national origin, age, abilities, ideas and perspectives, or socioeconomic status. Your suggestions are encouraged and appreciated. Feel free to let me know ways to improve the effectiveness of the course for you personally or for other students.

Course Structure

The course will be delivered via multiple methods including lectures, class discussion, presentations, and hands-on practice. For the majority of the class meetings, the class will start with a group discussion for the readings relating to the previous week topic & a lecture led by a fellow classmate for the first hour. The instructor will present the core topics for the second hour. The remaining time in the class will be hands-on practices related to the lecture topics covered or other in-class activities. Student participation is encouraged through reading discussions, hands-on practices, hands-on assignments, and class presentations.

Throughout the semester, hands-on practice and homework assignments provide students, including first-hand experience in spatial analysis using the software R and ArcGIS Pro. There are also reading assignments that help students evaluate and integrate the information acquired from the weekly readings. Students will design and conduct final projects to learn more in-depth spatial analysis approaches in a specific topic of research interest and demonstrate their ability to apply appropriate spatial analysis methods in resolving problems related to population, health and place.

This content is protected and may not be shared, uploaded, or distributed.

Course Content Distribution and Synchronous Session Recordings Policies

USC has policies that prohibit recording and distribution of any synchronous and asynchronous course content outside of the learning environment.
Recording a university class without the express permission of the instructor and announcement to the class, or unless conducted pursuant to an Office of Accessibility Services (OSAS) accommodation. Recording can inhibit free discussion in the future, and thus infringe on the academic freedom of other students as well as the instructor. (Living our Unifying Values: The USC Student Handbook, page 13).
Distribution or use of notes, recordings, exams, or other intellectual property, based on university classes or lectures without the express permission of the instructor for purposed other than individual or group study is prohibited. This includes but is not limited to providing materials for distribution by services publishing course materials. This restriction on unauthorized use also applies to all information, which has been distributed to students or in any way has been displayed for use in relationship to the class, whether obtained in class, via email, on the internet, or via any other media. (Living our Unifying Values: The USC Student Handbook, page 13).

Technological and Communication Requirements

The course will be focused on using R and ArcGIS Pro. Students can access the software for course assignments using computing resources provided by the Spatial Sciences Institute. Students will be given login credentials and instructions to sign-in to the remote Spatial Sciences Institute Server for the required software. Every student must have the following technology requirements:

- A computer with a fast Internet connection.
- A functional webcam and a microphone.
- An up-to-date web browser to access the Server

If a student does not have access to any of these, please speak with the instructor at the start of the semester. A limited number of computers with all the necessary software is available in the SSI Suite (AHF B55) during regular business hours, Monday to Friday, 9 a.m. to 5:00 p.m. These computers are available to any student in an SSCI or GSEC course and can be used as a resource if you experience difficulties in accessing the SSI server or using the GIS software on your personal computer.

SSI Server and Tech Support – This course utilizes the SSI Server which is a virtual desktop giving access to many different professional software. If you are unable to connect to the server or experience any type of technical issues, send an email to the instructor for the issues with descriptive information on the issues including error messages and screenshots, if available.

Communications – All materials to be handed in will be submitted via Brightspace. It is each student's responsibility to stay informed about what is going on in our course. Any important announcements will be posted on the News Announcements at the Brightspace Home page. Be sure to check the announcements each time you log onto Brightspace as well as set up automatic email notification of the announcements. Please be sure that you read as soon as

This content is protected and may not be shared, uploaded, or distributed.

possible all email sent from Brightspace or from me. Do not ignore course email until the day before assignments are due.

While I am usually on-line all day and will probably respond to emails from students very quickly, I will endeavor to respond to all email within 24 hours of receipt, aiming for no more than 72 hours delay. In the rare case when I expect to be off-line for more than 72 hours, I will post an announcement on the Blackboard site.

Required Readings and Supplementary Materials

The required textbooks for this course are:

- de Smith, Michael J., Michael F. Goodchild, and Paul A. Longley. 2025. *Geospatial analysis: a comprehensive guide to principles, techniques and software tools* (7th Ed.). Winchelsea, UK: Winchelsea Press. (Available online: http://www.spatialanalysisonline.com/index.html.)
- O'Sullivan, David, and David Unwin. 2014. *Geographic Information Analysis*, 2nd Edition. Hoboken, NJ: Wiley. (Available online via USC Libraries.)

Supplementary readings will be assigned from various sources and will be accessed via the USC Library's electronic collections and/or provided by the instructor via Blackboard, including:

- Adams, Marc A., Lawrence D. Frank, Jasper Schipperijn, Graham Smith, James Chapman, Lars B. Christiansen, Neil Coffee et al. 2014. "International variation in neighborhood walkability, transit, and recreation environments using geographic information systems: the IPEN adult study." International journal of health geographics 13, (1): 1-17.
- Almasad, A., Pavlak, G., Alquthami, T. and Kumara, S., 2023. Site suitability analysis for implementing solar PV power plants using GIS and fuzzy MCDM based approach. *Solar Energy*, 249, pp.642-650.
- Akinboyewa, T., Li, Z., Ning, H. and Lessani, M.N., 2025. GIS copilot: Towards an autonomous GIS agent for spatial analysis. *International Journal of Digital Earth*, 18(1), p.2497489.
- Baddeley, A., Nair, G., Rakshit, S., McSwiggan, G., & Davies, T. M. 2021. Analysing point patterns on networks—A review. *Spatial Statistics*, 42, 100435.
- Baker, David M., and Alain-Jacques Valleron. 2014. "An open source software for fast grid-based data-mining in spatial epidemiology (FGBASE)." *International journal of health geographics* 13, (1): 46.
- Barreto, Moises Colares, Doris Kosminsky, and Claudio Esperança. 2018. "Hexagonal hierarchical cartogram: Towards a thematic map of Brazil." InfoDesign-Revista Brasileira de Design da Informação 15 (1): 45-62.
- Benassi, F., Tomassini, C. and Di Felice, G., 2025. Spatial Heterogeneities or Inequalities? Health Care Supply and Demand of the Older Population in Italy. *Applied Spatial Analysis and Policy*, 18(1), p.44.

This content is protected and may not be shared, uploaded, or distributed.

- Burdick, K. J., Rees, C. A., Lee, L. K., Monuteaux, M. C., Mannix, R., Mills, D., Hirsh, M. P., & Fleegler, E. W. 2023. Racial & ethnic disparities in geographic access to critical care in the United States: A geographic information systems analysis. *PloS One*, *18*(11), e0287720–e0287720. https://doi.org/10.1371/journal.pone.0287720
- Debarsy, N. & Le Gallo, J. 2025. Identification of Spatial Spillovers: Do's and Don'ts. Journal of economic surveys: 39 (5), 2152–2173.
- Diriba, D., Karuppannan, S., Regasa, T. and Kasahun, M., 2024. Spatial analysis and mapping of malaria risk areas using geospatial technology in the case of Nekemte City, western Ethiopia. *International Journal of Health Geographics*, 23(1), p.27.
- Dutta, B., Das, M., Roy, U., Das, S., & Rath, S. 2022. Spatial analysis and modelling for primary healthcare site selection in Midnapore town, West Bengal. *GeoJournal,* 87(6), 4807-4836.
- Fränti, P., Sieranoja, S. and Laatikainen, T., 2025. Designing a clustering algorithm for optimizing health station locations. *International Journal of Health Geographics*, 24(1), p.4.
- Gibson, L., Engelbrecht, J., & Rush, D. 2019. Detecting historic informal settlement fires with sentinel 1 and 2 satellite data-Two case studies in Cape Town. Fire Safety *Journal*, *108*, 102828.
- Hanna-Attisha, Mona, Jenny LaChance, Richard Casey Sadler, and Allison Champney Schnepp. 2016. "Elevated blood lead levels in children associated with the Flint drinking water crisis: a spatial analysis of risk and public health response." *American Journal of Public Health* 106 (2): 283-290.
- Harris, Nancy L., Elizabeth Goldman, Christopher Gabris, Jon Nordling, Susan Minnemeyer, Stephen Ansari, Michael Lippmann et al. 2017. "Using spatial statistics to identify emerging hot spots of forest loss." *Environmental Research Letters* 12(2): 024012.
- Hetamsaria, P. & Matori, A.N. (2019). Applications of network analysis features in GIS. *Platform: A Journal of Engineering*, *3*(2), 46-54.
- Hohl, Alexander, Minrui Zheng, Wenwu Tang, Eric Delmelle, and Irene Casas. 2017. "Spatiotemporal point pattern analysis using Ripley's K function." In *Geospatial data science: techniques and applications*. CRC Press, Boca Raton, FL.
- Kelly-Schwartz, A. C., J. Stockard, S. Doyle, & M. Schlossberg. 2004. "Is sprawl unhealthy? A multilevel analysis of the relationship of metropolitan sprawl to the health of individuals." *Journal of Planning Education and Research*, 24(2), 184-196.

This content is protected and may not be shared, uploaded, or distributed.

- Kinsey, E.W., Neckerman, K.M., Quinn, J.W., Bader, M.D., Mooney, S.J., Lovasi, G.S., Kinsey, D. and Rundle, A.G., 2024. Using SaTScan to identify and map nightlife areas in Philadelphia, PA. Journal of maps, 20(1), p.2394505.
- Kopczewska, K., 2022. Spatial machine learning: new opportunities for regional science. *The Annals of Regional Science, 68(3), pp.713-755.*
- Kwan, M.P., 2022. Human mobility and the neighborhood effect averaging problem (NEAP). In *New thinking in GIScience* (pp. 95-101). Singapore: Springer Nature Singapore.
- Kwan, Mei-Po. 2018. "The limits of the neighborhood effect: Contextual uncertainties in geographic, environmental health, and social science research." *Annals of the American Association of Geographers* 108 (6): 1482-1490.
- Langford, Mitchel, Gary Higgs, and Richard Fry. 2016. "Multi-modal two-step floating catchment area analysis of primary health care accessibility." *Health & place* 38: 70-81.
- Li, M. and Ao, Y., 2026. Equity in public decision-making: a dynamic comparative study of urban–rural elderly care institution resource allocation in China. In *Institutional Elderly Care in China: Resource Allocation and Spatial Optimization* (pp. 81-112). Singapore: Springer Nature Singapore.
- Liu, H., Chen, W., Tang, J., Deng, M., Guo, Y. and Tang, Z., 2025. Revealing Urban Spatial Interaction Characteristics and Crowd Travel Patterns from Trajectory Data. *Annals of the American Association of Geographers*, 115(3), pp.559-577.
- Liu, S., Wu, A. M., & Ho, H. C. 2023. Spatial variability of diurnal temperature range and its associations with local climate zone, neighborhood environment and mortality in Los Angeles. *Urban Climate*, *49*, 101526.
- Liu, Qiannan, Zhiyun Ouyang, Ainong Li, and Weihua Xu. 2016. Spatial distribution characteristics of biomass and carbon storage in forest vegetation in Chongqing based on RS and GIS. *Nature Environment and Pollution Technology* 15 (4): 1381.
- Ma, Jing, Chunjiang Li, Mei-Po Kwan, and Yanwei Chai. 2018. “A Multilevel Analysis of Perceived Noise Pollution, Geographic Contexts and Mental Health in Beijing.” *International journal of environmental research and public health* 15, no. 7: 1479–.
- Mobley, Lee R., Tzy-Mey Kuo, Matthew Urato, Sujha Subramanian, Lisa Watson, and Luc Anselin. 2012. “Spatial heterogeneity in cancer control planning and cancer screening behavior." *Annals of the Association of American Geographers* 102 (5): 1113-1124.
- Morlighem, C., Nnanatu, C.C., Visée, C., Fall, A. and Linard, C., 2025. Spatial interpolation of health and demographic variables: Predicting malaria indicators with and without covariates. *PLoS One*, 20(5), p.e0322819.
- Mulrooney, Timothy, Kathi Beratan, Christopher McGinn, and Benjamin Branch. 2017. "A comparison of raster-based travel time surfaces against vector-based network calculations as applied in the study of rural food deserts." *Applied geography* 78: 12-21.
- Murray, A. T., Shammout, K., Farwell, R., Huetten, K., & Cox, T. 2024. Service area

This content is protected and may not be shared, uploaded, or distributed.

delineation to support on-demand mobility for the transport disadvantaged. *Socio-Economic Planning Sciences*, *95*, 101967-. https://doi.org/10.1016/j.seps.2024.101967

- Nelson, T., Frazier, A.E., Kedron, P., Dodge, S., Zhao, B., Goodchild, M., Murray, A., Battersby, S., Bennett, L., Blanford, J.I. and Cabrera-Arnau, C., 2025. A research agenda for GIScience in a time of disruptions. *International Journal of Geographical Information Science*, 39(1), pp.1-24.
- Okundi, A.O. and Varol, C., 2024. Spatial analysis of primary healthcare accessibility patterns in Migori County, Kenya. SSM-Health Systems, 2, p.100005.
- Perchoux, Camille, Basile Chaix, Ruben Brondeel, and Yan Kestens. 2016. "Residential buffer, perceived neighborhood, and individual activity space: New refinements in the definition of exposure areas–The RECORD Cohort Study." *Health and Place* 40: 116-122.
- Rong, C., Ding, J. and Li, Y., 2024. An interdisciplinary survey on origin-destination flows modeling: Theory and techniques. *ACM Computing Surveys*, 57(1), pp.1-49.
- Spencer, John, and Gustavo Angeles. 2007. "Kernel density estimation as a technique for assessing availability of health services in Nicaragua." *Health Services and Outcomes Research Methodology* 7: 145-157.
- Villarreal, M.L., Bishop, T.B., Sankey, T.T., Smith, W.K., Burgess, M.A., Caughlin, T.T., Gillan, J.K., Havrilla, C.A., Huang, T., LeBeau, R.L. and Norton, C.L., 2025. Applications of unoccupied aerial systems (UAS) in landscape ecology: a review of recent research, challenges and emerging opportunities. *Landscape Ecology*, 40(2), p.43.
- Wei, R., Rey, S., & Knaap, E. 2021. Efficient regionalization for spatially explicit neighborhood delineation. *International Journal of Geographical Information Science*, 35(1), 135–151.
- Weng, Q., Li, Z., Cao, Y., Lu, X., Gamba, P., Zhu, X., Xu, Y., Zhang, F., Qin, R., Yang, M.Y. and Ma, P., 2024. How will ai transform urban observing, sensing, imaging, and mapping?. *npj Urban Sustainability*, 4(1), p.50.
- Yeager, R., Browning, M.H., Breyer, E., Ossola, A., Larson, L.R., Riggs, D.W., Rigolon, A., Chandler, C., Fleischer, D., Keith, R. and Walker, K., 2023. Greenness and equity: Complex connections between intra-neighborhood contexts and residential tree planting implementation. *Environment international*, 176, p.107955.
- Zhang, Y., Mavoa, S., Zhao, J., Raphael, D., & Smith, M. 2020. The association between green space and adolescents' mental well-being: a systematic review. *International journal of environmental research and public health*, 17(18), 6640.

This content is protected and may not be shared, uploaded, or distributed.

Description and Assessment of Assignments

There are several different types of assignments due nearly each week in our course that make up a significant part of your final grade. These different types of assignments are described below.

Lead Class Discussion - 1 worth a total of 10%: Each student will lead one class discussion on the assigned readings once during the semester. Class discussion will focus on the theory and application of the methods as presented in the weekly readings. Each student discussion leader prepares for a 15-minute opening statement presentation followed by 3 questions raised from the presentation material for class discussion.

Written Reflection on Reading/Class Discussion – A total of 14%. Students are expected to read all material outlined for each week before class, participating in in-class discussion led by a fellow classmate, and reflect on the material via online forum. The number of the written essays/reflections depends on the number of enrollment and therefore the number of lead class discussion that semester. This written reflection aims to help the students evaluate and integrate the information acquired from the course readings and in-class discussion.

In-Class Engagement – a total of 12%. An in-class engagement grade for the semester will be assigned based upon student engagement in the class sessions including, but not limited to, in-class exercises, participations to in-class discussion, and feedback to student project ideas. Failure to attend, participate in class exercise, or not be adequately prepared to discuss the readings will lead to the assignment of a lower grade for that week.

Hands-on Assignments -- 6 worth a total of 24%: Students will be assigned a total of 6 hands-on assignments in this course. These assignments are to practice spatial analysis techniques explored in theory in the texts. You will analyze the data using spatial analysis tools in ArcGIS, R and/or GeoDa, and write a short report to answer the questions in each assignment. When class time allows besides regular lectures and discussions, we might work on some exercises in the assignments during the class, but it is student's responsibility to complete the assignments by the stated deadlines outside the classroom time.

Final Project

To integrate your learning of all the material covered in the course, in the final project, each student will conduct an independent research project. You will design, undertake, and report on a spatial analysis research project that will be the context of different assignments throughout the semester. The four components of the project will be due at different times throughout the semester. The grading rubric for each project component will be provided at the time the assignment is released. The four final project components include:

Project Idea Presentation - 4%. An in-class presentation for your final project idea, including the spatial questions you would like to investigate, how you plan to solve them, the data to be used for analysis, and expected outcomes.

Proposal – 10%. A proposal document developed from the project idea presentation, your background research, and the feedback received.

This content is protected and may not be shared, uploaded, or distributed.

Final Presentation - 10%. A final presentation during the final week of the class session.

Report - 18%. A written report in the format of a research paper on your final project methodology and outcomes.

Grading Breakdown

Assessments	Number	% Each	Total % of Grade
In-class engagement	-	-	12
Lead class discussion	1	10	10
Written reflection for class reading/discussion	6	2	12
Hands-on assignments	6	4	24
Project idea presentation	1	4	4
Project proposal	1	10	10
Project presentation	1	10	10
Project report	1	18	18
Total	-	-	100

Assignment Submission Policy

Unless otherwise noted, assignments must be submitted via Blackboard by the due dates specified in the Course Schedule below and on the assignment instructions.

Students are expected to attend and participate in every class session and to complete and upload all assignments before the deadlines detailed in the Course Schedule. Late work will be assessed a penalty of 10% per day and zero grades will be assigned for work that is more than one week late.

Course Schedule: A Weekly Breakdown

Modules

1	Conceptual framework for spatial data analysis
2	Spatial patterns and dependencies
3	Modeling and mapping continuous space processes
4	Spatial connectivity and accessibility

		Topics	Readings & Assignments	Deliverables
	Week 1 1/16	Introduction to course & spatial analysis conceptual framework	de Smith et al. (2025) Ch. 2 O'Sullivan & Unwin (2014) Ch.1 ~~Perchoux~~ et al. (2016) Nelson et al. (2025)	

This content is protected and may not be shared, uploaded, or distributed.

	Week 2 1/23* *Monday, 1/19 is university holiday (MLK Day)	Geospatial data analysis pitfalls and opportunities	O'Sullivan & Unwin (2014) Ch.2 Wei et al. (2021) Kwan (2018) Kwan (2022) **Assignment 1**	Sign-up for class presentations
	Week 3 1/30	Maps and visualization for spatial data (+ Guest talk on spatial data resources: Andy Rutkowski)	O'Sullivan & Unwin (2014) Ch.3 & 4 Barreto et al. (2018) Mobley et al. (2012)	
	Week 4 2/6	Spatial point patterns: density-based and distance-based methods	de Smith et al. (2025). Ch. 5-§5.2 & §5.4 Baddeley et al. (2021) Hohl et al. (2017) **Assignment 2**	Assignment 1 due on Monday, 2/2
	Week 5 2/13	Spatial clustering: spatial autocorrelation, local statistics and spatial scan statistics	O'Sullivan & Unwin (2014) Ch.7 & 8 Harris et al. (2017) Li et al. (2025) Kinsey et al. (2024)	
	Week 6 2/20* *Monday, 2/16 is university holiday (President's Day)	Spatial spillovers and contextual effects: spatial autoregressive models	Yin et al. (2018) Wang et al. (2022) Debarsy and Le Gallo (2025) **Assignment 3**	Assignment 2 due on Tuesday, 2/17 Project Idea Presentation (no discussion)
	Week 7 2/27	Concepts and processes of spatial fields	O'Sullivan & Unwin (2014) §9.4-9.5 Baker and Valleron (2014) Mulrooney et al. (2017)	
	Week 8 3/6	Spatial decision modeling on fields (+ Guest research talk: TBA)	Almasad et al. (2023) Dutta et al. (2022) Diriba et al. (2024) **Assignment 4**	Assignment 3 due on Monday, 3/2

This content is protected and may not be shared, uploaded, or distributed.

	Week 9 3/13	Variogram & spatial interpolations	O'Sullivan &Unwin (2014) §10.3-§10.4 Hanna-Attisha et al. (2016) Morlighem et al. (2025)	Project Proposal: due Friday, 3/13
	** 3/15-3/22 is Spring Recess*			
	Week 10 3/27	Remote sensing as spatial field measurement	Liu et al. (2023) Yeager et al. (2023) Villarreal et al. (2025) **Assignment 5**	Assignment 4 due on Monday, 3/23
	Week 11 4/3	Network concept and optimization of spatial services	Adams et al. (2014) Hetamsaria & Matori (2019) Franti et al. (2025)	
	Week 12 4/10	Network-based accessibility for public health and environmental equity	Burdick et al. (2023) Murray (2024) Okundi et al. (2024) **Assignment 6**	Assignment 5 due on Monday, 4/6
	Week 13 4/17	Spatial interaction and flow processes	Benassi et al. (2025) Rong et al. (2023) Liu et al. (2025)	
	Week 14 4/24	AI for spatial analysis	Kopczewska (2022) Akinboyewa et al. (2025) Weng et al. (2024)	Assignment 6 due on Monday, 4/20
	Week 15 5/1* *Friday, 5/1 is the last day of class	Final Presentations		Final presentation during the class
	Final Exam Week	**Final report due on Sunday, May 10th**		

Statement on Academic Conduct and Support Systems

Academic Integrity

The University of Southern California is a learning community committed to developing successful scholars and researchers dedicated to the pursuit of knowledge and the dissemination of ideas.

This content is protected and may not be shared, uploaded, or distributed.

Academic misconduct, which includes any act of dishonesty in the production or submission of academic work, compromises the integrity of the person who commits the act and can impugn the perceived integrity of the entire university community. It stands in opposition to the university's mission to research, educate, and contribute productively to our community and the world.

All students are expected to submit assignments that represent their own original work, and that have been prepared specifically for the course or section for which they have been submitted. You may not submit work written by others or "recycle" work prepared for other courses without obtaining written permission from the instructor(s).

Other violations of academic integrity include, but are not limited to, cheating, plagiarism, fabrication (e.g., falsifying data), collusion, knowingly assisting others in acts of academic dishonesty, and any act that gains or is intended to gain an unfair academic advantage.

The impact of academic dishonesty is far-reaching and is considered a serious offense against the university. All incidences of academic misconduct will be reported to the Office of Academic Integrity and could result in outcomes such as failure on the assignment, failure in the course, suspension, or even expulsion from the university.

For more information about academic integrity see the Student Handbook or the Office of Academic Integrity's website, and university policies on Research and Scholarship Misconduct. Please ask your instructor if you are unsure what constitutes unauthorized assistance on an exam or assignment, or what information requires citation and/or attribution.

Students and Disability Accommodations:

USC welcomes students with disabilities into all of the University's educational programs. The Office of Student Accessibility Services (OSAS) is responsible for the determination of appropriate accommodations for students who encounter disability-related barriers. Once a student has completed the OSAS process (registration, initial appointment, and submitted documentation) and accommodations are determined to be reasonable and appropriate, a Letter of Accommodation (LOA) will be available to generate for each course. The LOA must be given to each course instructor by the student and followed up with a discussion. This should be done as early in the semester as possible as accommodations are not retroactive. More information can be found at osas.usc.edu. You may contact OSAS at (213) 740-0776 or via email at osasfrontdesk@usc.edu.

Support Systems:

Counseling and Mental Health - (213) 740-9355 – 24/7 on call
Free *and* confidential mental health treatment for students, including short-term psychotherapy, group counseling, stress fitness workshops, and crisis intervention. *988 Suicide and Crisis Lifeline* - 988 *for both calls and text messages – 24/7 on call*
The 988 Suicide and Crisis Lifeline (formerly known as the National Suicide Prevention Lifeline) provides free and confidential emotional support to people in suicidal crisis or emotional

This content is protected and may not be shared, uploaded, or distributed.

distress 24 hours a day, 7 days a week, across the United States. The Lifeline is comprised of a national network of over 200 local crisis centers, combining custom local care and resources with national standards and best practices. The new, shorter phone number makes it easier for people to remember and access mental health crisis services (though the previous 1 (800) 273-8255 number will continue to function indefinitely) and represents a continued commitment to those in crisis.

CARE-SC - (213) 740-9355(WELL) – 24/7 on call

Free and confidential therapy services, workshops, and training for situations related to gender- and power-based harm (including sexual assault, intimate partner violence, and stalking).

Office for Civil Rights Compliance - (213) 740-5086

Information about how to get help or help someone affected by harassment or discrimination, rights of protected classes, reporting options, and *additional* resources for students, faculty, staff, visitors, and applicants.

The Office of Student Accessibility Services (OSAS) - (213) 740-0776

OSAS ensures equal access for students with disabilities through providing academic accommodations and auxiliary aids in accordance with federal laws and university policy.

USC Campus Support and Intervention - (213) 740-0411

Assists students and families in resolving complex personal, financial, and academic issues adversely affecting their success as a student.

USC Culture Journey - (213) 740-8258

At USC our six unifying values support our University's mission, and guide our behaviors, the way we engage with ourselves and our community, to create our Trojan culture.

USC Emergency - UPC: (213) 740-4321, HSC: (323) 442-1000 – 24/7 on call

Emergency assistance and avenue to report a crime. Latest updates regarding safety, including ways in which instruction will be continued if an officially declared emergency makes travel to campus infeasible.

USC Department of Public Safety - UPC: (213) 740-6000, HSC: (323) 442-1200 – 24/7 on call

Non-emergency assistance or information.

Office of the Ombuds - (213) 821-9556 (UPC) / (323-442-0382 (HSC)

A safe and confidential place to share your USC-related issues with a University Ombuds who will work with you to explore options or paths to manage your concern.

Occupational Therapy Faculty Practice - (323) 442-2850 or otfp@med.usc.edu

Confidential Lifestyle Redesign services for USC students to support health promoting habits and routines that enhance quality of life and academic performance.

This content is protected and may not be shared, uploaded, or distributed.